AI-Powered Delivery Exception Resolution

Project 3 | Low-Code Multi-Agent System

25 September 2026

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Executive Summary

The completed notebook was executed against all attached data.

SCOPE

13 raw delivery-log rows
10 unique shipments
12 customers + 6 lockers
10-page operations playbook

RECOMMENDATION

Keep deterministic policy rules authoritative for safety and escalation. Use the LLM for classificat

RESULT

Task completion: 100%
Escalation accuracy: 100% (8/8)
Tool-call accuracy: 100%
Coherence: 5.00 / 5 (offline trace checklist)

- The system filters routine noise before unnecessary tool or model calls.
- It removes duplicate scans, retrieves policy context, checks lockers, and applies escalation rules.
- Structured Resolution and Communication agents are validated by critic agents before finalization.
- The run used a credential-free deterministic fallback; remote GPT-4o-mini mode remains supported when configured.

Business Problem Overview and Solution Approach

SOLUTION

A hub-and-spoke LangGraph workflow combines guardrails, deterministic tools, structured agent outputs,

scans create false work, natural-language notes hide custom

- 1. Preprocess: deduplicate, consolidate attempts, scan for injection, and filter routine noise.
- 2. Enrich: retrieve customer profile, locker availability, playbook chunks, and escalation signals.
- 3. Decide: Resolution Agent selects classification and action with a policy-grounded rationale.
- 4. Validate: Resolution Critic may accept, revise, or escalate; Communication Critic validates the message.
- 5. Finalize: package an auditable action, escalation flag, customer message, citations, and latency.

Data Loading

Source coverage and attributes used by the solution

WS

us_code, driver description, customer_id, address, package type/s

CUSTOMER DB | 12 ROWS

Customer tier, preferred channel, exception count in the last 90 days, active credit, and name. Nam

status, operating hours, and max package size. Eligibility

PLAYBOOK + LABELS

10-page PDF split into 24 page-aware chunks

Ground truth has 13 labels aligned to the raw logs and consolid

- Important data quality handling: duplicate scans for SHP-002 and SHP-007 are removed before consolidation.
- Ground truth is consolidated by shipment, using the highest-attempt exception label for multi-row shipments.

Model Initialization and Retrieval

MOTE MODE

deterministic offline fallback because no credentials were attached. This ma and still

dation: gpt-4o-mini

nperature: 0.0

dentials loaded only from config.json or env

RETRIEVAL

ChromaDB cosine search over 24 local TF-IDF vectors returns ranked policy chunks with source

- Pydantic schemas enforce valid Resolution, Communication, and Critic outputs.
- Temperature 0.0 is used to reduce variance in structured decisions.
- The notebook supports remote LLM execution when a valid config.json is supplied, but never hard-codes credentials.
- The offline path is explicitly labeled in the exported results as runtime_mode=offline_fallback.

Defining Tools

Delivery Log Reader
Reads all raw CSV rows and feeds the preprocessor.

Customer Profile Lookup
Parameterized SQLite query; PII name is returned only with explicit include_pii=True.

Locker Availability Checker
Checks ZIP, size capacity, FULL/LIMITED status, and returns eligibility plus reason.

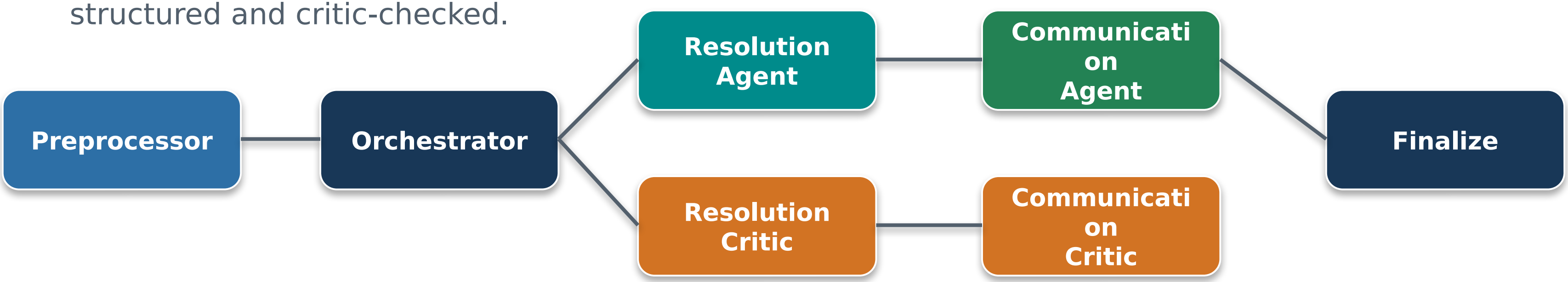
Playbook Search
Retrieves relevant page-aware policy chunks for the Resolution and Critic agents.

Escalation Rule Engine
Deterministically evaluates automatic and discretionary triggers, including delay-hour parsing.

- Evaluation result: 10/10 scenarios invoked the expected tool path, including the intended noise short-circuit.

Multi-Agent System Architecture

Hub-and-spoke orchestration keeps routing deterministic while allowing agent outputs to be structured and critic-checked.



Preprocessor

Orchestrator

Resolution Agent

Communication Agent

Resolution Critic

Communication Critic

Finalize

- Prompts: policy-grounded resolution, tier-aware communication, resolution validation, communication validation, and 1–5 trajectory coherence judging.
- Escalation: automatic triggers (third attempt, VIP ≥3 exceptions, damaged perishables, suspected fraud, safety) plus discretionary triggers (high-history Standard and delayed Premium perishables).
- Security: parameterized SQL, default PII redaction, prompt-injection scanning before LLM calls, no raw PII in coherence judging, and safe fallback escalation.

Test Cases Execution, Evaluation, and Observability

VABILITY

eption trace records tool [illegible]

AGGREGATE METRICS

Task completion 100%
Escalation accuracy 100% (8/8)
Tool-call accuracy 100%
Coherence 5.00/5 (offline checklist

ACCURACY COMMENT

All expected decisions, exception classifications, and escalation decisions match the

- SHP-001 Delivered → noise / no action
- SHP-002 VIP attempts → reschedule / escalate
- SHP-003 Address issue → reschedule
- SHP-004 VIP fragile damage → replace / escalate
- SHP-005 Third attempt → locker reroute / escalate
- SHP-006 Refused → return to sender
- SHP-007 Perishable weather >4h → replace / escalate
- SHP-008 Standard high history → reschedule / escalate
- SHP-009 Damaged perishable → replace / escalate
- SHP-010 Depot scan → noise / no action

APPENDIX

Detailed scenario outcomes, citations, and submission checklist

Appendix: Detailed Evaluation Results

Shipment	Outcome / tone	Esc.	Task	Tools	Coherence	Observation	Playbook
SHP-001	NO EXCEPTION / —	N/A	PASS	PASS	5/5	Delivered scan filtered as routine noise.	N/A
SHP-002	RESCHEDULE / formal	PASS	PASS	PASS	5/5	Repeat-attempt VIP: reschedule and supervisor review.	P2, 5, 6
SHP-003	RESCHEDULE / casual	PASS	PASS	PASS	5/5	Address mismatch: correct details, then retry.	P3, 2, 3
SHP-004	REPLACE / formal	PASS	PASS	PASS	5/5	VIP fragile damage: replace and escalate.	P4, 4, 7
SHP-005	LOCKER REROUTE / casual	PASS	PASS	PASS	5/5	Third attempt: eligible locker route and escalate.	P2, 2, 3
SHP-006	RETURN / casual	PASS	PASS	PASS	5/5	Refused delivery: return to sender; no escalation.	P5, 5, 4
SHP-007	REPLACE / formal	PASS	PASS	PASS	5/5	Perishable weather delay >4 h: replace and escalate.	P6, 2
SHP-008	RESCHEDULE / casual	PASS	PASS	PASS	5/5	High-history Standard customer: discretionary escalation.	P2, 5, 5
SHP-009	REPLACE / formal	PASS	PASS	PASS	5/5	Damaged perishable: replace and escalate.	P4, 6, 2
SHP-010	NO EXCEPTION / —	N/A	PASS	PASS	5/5	Depot scan filtered as operational noise.	N/A

Escalation is N/A for the two noise cases. Coherence 5/5 uses the offline trace-completeness checklist, not an independent LLM judge; mean 0.01s latency is local fallback time, not remote-model or production latency.

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Conclusions and Business Recommendations

NEXT PRODUCTION STEPS

Persist shipment notes and supervisor queues, add temperature-controlled locker metadata, and expand a

. Duplicate scans and noise were filtered; policy-based escal

Power Ahead!

- Keep deterministic policy rules authoritative for safety and escalation.
- Use the LLM for flexible language tasks: classification rationale and customer-facing communication.
- Monitor task completion, escalation accuracy, tool-call accuracy, coherence, and latency in production.
- Keep the offline fallback as a reproducible evaluation and disaster-recovery path.